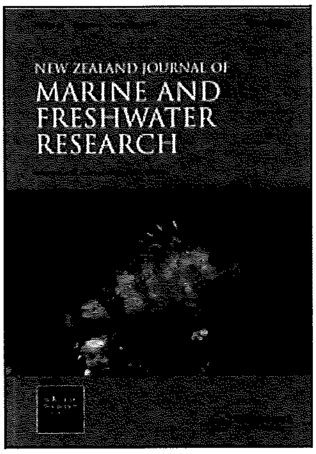

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Thermal tolerance and preference of some native New Zealand freshwater fish

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Thermal tolerance and preference of some native New Zealand freshwater fish

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Abstract Upper lethal and preferred temperatures were determined experimentally for eight common species of New Zealand freshwater fish. Upper lethal temperatures ranged between 28.3 and 39.7°C and preferred temperatures between 16.1 and 26.9°C. *Anguilla australis* was the most tolerant of high temperatures, whereas *Retropinna retropinna* and two species of *Galaxias* preferred cooler water. The relationship between lethal and preferred temperatures for New Zealand species was similar to that calculated from data for 38 other species of freshwater fish, meaning that either lethal or preferred temperatures could be used accurately to predict lethal, preferred, and growth optima, and to recommend temperature regimes for waterways. However, New Zealand field records show that significant relationships between fish density and water temperature exist for only two species, *A. dieffenbachii* and *Cheimarrichthys fosteri*. This suggests New Zealand species are able to thrive within a wide temperature range.

Keywords preferred temperature; freshwater fish; lethal temperature; New Zealand

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INTRODUCTION

The importance of temperature as a factor that controls the distribution, abundance, physiology, and behaviour of fish is well established (Alabaster & Lloyd 1982; Coutant 1987). Fish are extremely sensitive to temperature and will select those temperatures where physiological functions operate at maximum efficiency (Crawshaw 1977). While fish can survive, within limits, in temperatures outside their optimal ranges, resulting physiological or behavioral changes can decrease their chances of survival and reproductive success (Reynolds 1977).

Concern about the effects of heated effluents on native freshwater fish prompted this study. While there are many references to the lethal, avoided, and preferred temperatures of fish from North America and Europe (e.g., Coutant 1977; Jobling 1981; Alabaster & Lloyd 1982), with the exception of Boubée et al. (1991), in New Zealand, only lethal levels have been determined and the results are presented in unpublished theses (Jellyman 1974; Teale 1986; Main 1988) or informal reports (Simons 1984, 1986). The primary objective of this project was to determine the lethal and preferred temperatures of common native freshwater fish species and to compare the results with other known thermal information. A secondary objective was to compare the relationships between lethal and preferred temperatures described by Jobling (1981) with results for New Zealand species, to determine if these could be used successfully by local water managers, together with the guidelines issued by Armour (1991), to recommend temperature regimes for freshwater fish in New Zealand waterways.

METHODS

Collection and acclimation of animals

The thermal tolerance and preference of eight common native freshwater fish were determined. *Galaxias maculatus* (inanga), *G. fasciatus* (banded kokopu), *Anguilla australis* (shortfinned eel), *A. dieffenbachii* (longfinned eel), *Retropinna retropinna* (common smelt), *Gobiomorphus cotidianus* (common bully), *G. basalis* (Cran's bully), and *Cheimarrichthys fosteri* (torrentfish), were collected from various locations in the North Island and acclimated at 15°C for at least 24 h prior to testing.

Before acclimation, all fish were treated with 2 mg l^{-1} furanace-p to reduce the risk of disease. De-chlorinated, aerated tap water was used for acclimation and testing. Fish were kept on a 12:12 photoperiod and the water was aerated continuously during acclimation and testing.

Temperature tolerance

Although a number of approaches for determining the upper temperature tolerances of fish are described in the literature (Fry 1967, 1971), the following two methods have most commonly been employed. In the first method, the temperature is increased at a constant rate, starting from the acclimation temperature, until the fish become narcotised and fail to respond to stimuli. This incapacitated state is referred to as the end point or "critical thermal maximum" (CTM) (Becker & Genoway 1979; Paladino et al. 1980). The temperature by which half the fish being tested have succumbed is the CTM for that species. This has been the most commonly used method in New Zealand (Simons 1984, 1986).

The other commonly used method is to abruptly transfer the fish to water which has already been heated to a specific temperature, and to record the response of the fish over a given time period, usually 96 hours (Fry 1971). By using different temperatures, the temperature which is lethal to fish can be determined. We chose the second method, as it mimics more closely the situation of a fish being subject to an abrupt change in temperature in the wild, such as might be encountered below a heated effluent discharge. We used a relatively short exposure period of 10 minutes as this reflects the time required for a rapidly mixed discharge to integrate with the receiving waters (Keller 1993).

For these tests, fish were placed in polyethylene containers (170 × 170 × 200 mm) which had 100 × 80 mm windows of 2 mm mesh 20 mm above the floor on two sides. These containers allowed transfer of fish between water baths without handling and were used during acclimation and testing. Three replicate containers, each containing 3 to 5 fish, were immersed in an aerated temperature controlled bath for 10 minutes. The test temperatures started at 20±0.2°C and continued at 2 to 5°C intervals until 100% deaths occurred. Fish were considered to be dead if they were drifting ventral side up, and failed to recover equilibrium or show signs of breathing when returned to the acclimation temperature.

As a control, three replicate containers of fish were transferred to a water bath at the acclimation temperature for 10 minutes to check that the transfer process itself was not lethal. As a further check of the health of the fish, three replicate containers were maintained without any disturbance for the length of the experiment. If 10% or more of either group of control fish died, the test was declared invalid.

The percentage of total deaths was plotted against the test temperature for each species. The LT_{50} for each species was determined by visually interpolating the temperature at which 50 percent deaths occurred.

Temperature preference

Because most New Zealand freshwater fish show a positive rheotactic response (attraction to flowing water), a thermal gradient tank was designed in which the water remained static. A perspex tank (2400 × 200 × 200 mm) was constructed with a series of six pipes running lengthwise along the bottom. Connections to heating and cooling units allowed hot water to be pumped through two pipes, while cold water was pumped in the opposite direction through the other four pipes. The temperatures of the hot (35 to 55°C) and cold (1.5 to 5°C) water supplies were altered to create different thermal gradients along the length of the tank. A maximum gradient of about 10°C could be obtained within the ranges tested (9 to 35°C).

Nylon mesh (1 mm mesh) laid along the floor and back wall of the tank prevented contact by fish with the heating or cooling pipes and made the subjects easier to locate. Along the back wall behind the mesh, a length of diffusion air tubing aerated the water and created enough circulation to prevent vertical thermal stratification. Baffles placed under the mesh at 200 mm intervals helped prevent longitudinal mixing.

The thermal gradient tank was placed in a dark room at constant (15°C) temperature. "Growlux" florescent tubes positioned above the tank provided even and continuous light over the entire tank; these lights remained on during testing. A thermal gradient was established in the tank prior to introducing the fish. The gradient chosen depended on the species and life stage being tested; when the gradient was outside the preferred temperature range, the fish tended to congregate at one end of the tank. The temperature gradient was adjusted so that the fish spent most of their time towards the middle of the tank.

Between 5 to 22 fish were placed in the gradient tank initially at the position where the temperature was as close as possible to the acclimation temperature (15°C) and left to settle for 2 to 3 hours before observations started. The experiments were run for 2 to 5 days, and 3 to 4 trials were conducted for each species or life stage. The experimental room was not entered during the experiment; all alterations and observations were made from outside the room.

Horizontal temperature profiles of the tank were obtained at the beginning and end of each experiment. In addition, digital thermometers set at each end of the tank indicated temperatures during the experiments. The hot and cold temperatures could be adjusted during the experiment to give different ranges in temperature and thus check that the observed preference location was not a preference for a particular part of the tank.

Experiments were monitored using four high resolution black and white video cameras connected through a sequential switcher to a monitor. Each camera's field of view covered a quarter (600 mm) of the gradient tank. The positions of the fish were recorded on a time-lapse video recorder such that 72 hours of observation were condensed onto a standard three hour tape.

The video tapes were used to count the number of fish in each marked 200 mm section of the gradient tank at 15 minute intervals. For the whole experimental period, the number of fish in each of the 12 sections was summed and converted to percentages. Results from individual trials were combined, and the cumulative percentage graphed against the temperatures which corresponded to each 200 mm marked section. The median and quartile values were visually interpolated; the median value represented the preferred temperature for that species.

Relationship between lethal and preferred temperatures

Jobling (1981) showed there was a linear relationship between lethal and preferred temperatures for fish. To test the applicability of Jobling's relationships to New Zealand species, we added the New Zealand data to that of Jobling, re-calculated the regression equation, and compared this to Jobling's original equation. We also calculated and compared a regression equation for just the New Zealand species as a further test of independence.

RESULTS

Data from the upper lethal (Fig. 1) and preferred temperature tests (Fig. 2) for two species, *R. retropinna* and *A. australis*, are shown as examples of species specific curves and data interpretation. Upper lethal and preferred temperature values for the eight species tested, together with data from other sources, including four additional species (*Gobiomorphus breviceps* (upland bully), *Galaxias argenteus* (giant kokopu), *G. postvectis* (shortjawed kokopu), and *G. brevipinnis* (koaro)) are presented in Table 1. *A. australis* was tolerant of the highest temperatures, whereas *G. brevipinnis* was the most sensitive.

Addition of data from New Zealand species made little difference to Jobling's (1981) regression equation between lethal and preferred temperatures (Table 2). Most New Zealand species had higher than expected lethal temperatures compared to their preferred temperatures (Fig. 3), but the regression equation for just the New Zealand species was also significant ($P < 0.01$) with a high correlation coefficient (Table 2).

DISCUSSION

Hokanson (1977) described fish with an upper lethal temperature between 28 and 34°C, or physiological optimum between 20 and 28°C, as mesotherms; these criteria apply to most of the New Zealand species tested. Further comparisons are difficult as nine of the species listed in Table 1 are endemic, and for those species more widespread in the Southern Hemisphere (*A. australis*, *G. maculatus*, and *G. brevipinnis*), little published material exists about their thermal tolerances.

Within the published New Zealand data, there are problems with the use of different acclimation temperatures and different test methods. However, where LT_{50} and CTM are compared for the same life stage and acclimation temperature, e.g., *G. fasciatus* whitebait at 16°C or adult *R. retropinna* at 20°C, there is good agreement between the

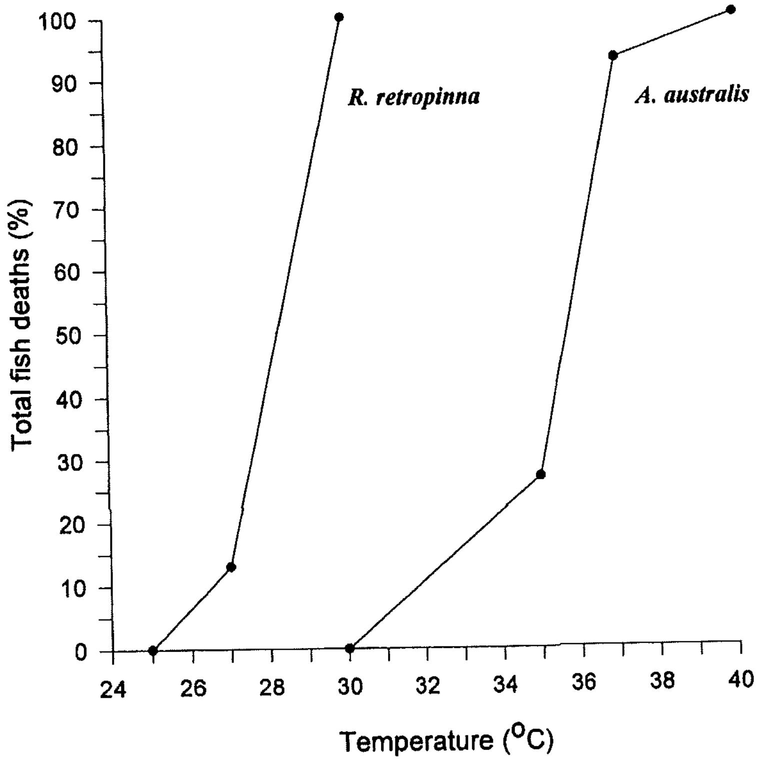

Fig. 1 Examples of plots from the lethal temperature tests for *R. retropinna* (n = 15) and *A. australis* elvers (n = 15).

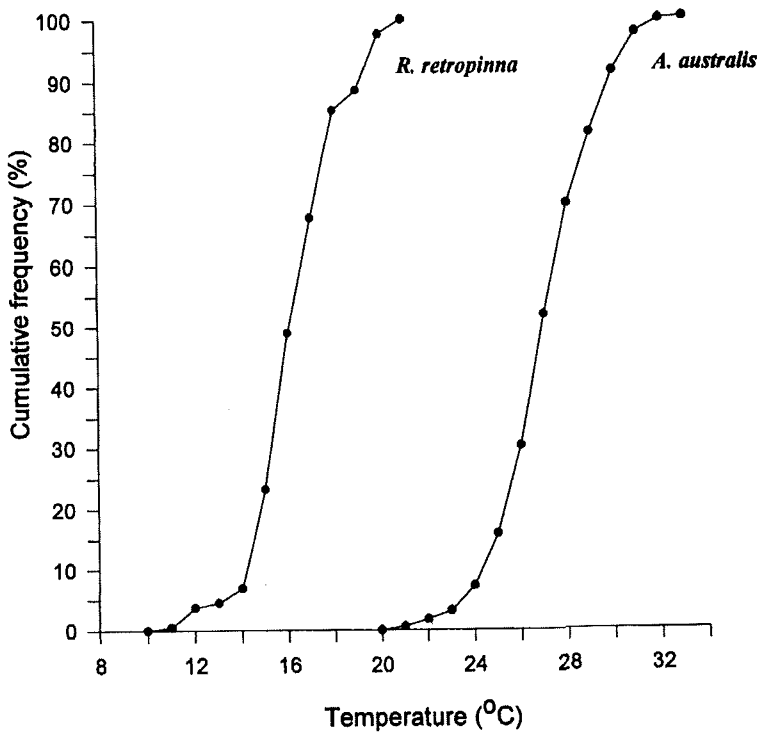

Fig. 2 Examples of plots from the temperature preference tests for *R. retropinna* (4 trials, total n = 21) and *A. australis* elvers (3 trials, total n = 36).

Table 1 Upper lethal and preferred temperatures for 12 New Zealand freshwater fish. (* = not determined).

Species	Life stage	Acclimated temp. (°C)	Upper lethal temp. (°C) and method		Preferred temp. and quartiles (°C)	Source
Anguilla australis (Shortfinned eel)	Glass eel	15	28.0	LT_{50}	–*	Jellyman 1974
	Elver	12	33.4	CTM	–	Simons 1986
		15	35.7	LT_{50}	26.9 (25.6–28.5)	this study
		20	35.9	CTM	–	Simons 1986
		21.5	36.0	CTM	–	Simons 1986
		22	36.3	CTM	–	Simons 1986
		25	37.3	CTM	–	Simons 1986
		28	38.1	CTM	–	Simons 1986
		30	30.5	CTM	–	Simons 1986
	Adult	15	39.7	LT_{50}	–	this study
A. dieffenbachii (Longfinned eel)	Glass eel	15	25.0	LT_{50}	–	Jellyman 1974
	Elver	15	34.8	LT_{50}	24.4 (22.6–26.2)	this study
	Adult	15	37.3	LT_{50}	–	this study
Gobiomorphus basalis (Cran's bully)	mixture	12	32.3	CTM	–	Simons 1984
		15	30.9	LT_{50}	21.0 (19.6–22.1)	this study
		20	33.9	CTM	–	Simons 1984
G. cotidianus (Common bully)	mixture	12	32.7	CTM	–	Simons 1984
		15	30.9	LT_{50}	20.2 (18.7–21.8)	this study
		20	34.0	CTM	–	Simons 1984
G. breviceps (Upland bully)	Juvenile	15	32.8	CTM	–	Teale 1986
Cheimarrichthys fosteri (Torrentfish)	Adult	15	30.0	LT_{50}	21.8 (20.1–22.9)	this study
Galaxias maculatus (Inanga)	Whitebait	15	–	–	18.8 (18.0–19.8)	this study
		20	33.1	CTM	–	Simons 1986
	Juvenile	12	30.5	CTM	–	Simons 1986
		15	–	–	18.7 (17.3–20.0)	this study
		16	31.7	CTM		Simons 1986
		20	32.9	CTM	–	Simons 1986
		22	33.8	CTM	–	Simons 1986
		26.5	35.4	CTM	–	Simons 1986
	Adult	15	30.8	LT_{50}	18.1 (17.2–19.1)	this study
G. argenteus (Giant kokopu)	Whitebait	16	30.0	CTM	–	Main 1988
		16	29.0	LT_{50}	–	Main 1988
G. postvectis (Shortjawed kokopu)	Juvenile	16	30.0	CTM	–	Main 1988
		16	29.0	LT_{50}	–	Main 1988
G. fasciatus (Banded kokopu)	Whitebait	14	30.6	CTM	–	Simons 1986
		15	–	–	16.1 (14.8–17.7)	this study
		16	30.0	CTM	–	Main 1988
		16	29.0	LT_{50}	–	Main 1988
		20	32.5	CTM	–	Simons 1986
		22	31.2	CTM	–	Simons 1986
		24	34.0	CTM	–	Simons 1986
		26	31.1	CTM	–	Simons 1986
	Adult	15	28.5	LT_{50}	17.3 (16.3–18.3)	this study
G. brevipinnis (Koaro)	Juvenile	16	28.0	CTM	–	Main 1988
		16	27.0	LT_{50}	–	Main 1988
Retropinna retropinna (Common smelt)	Adult	15	28.3	LT_{50}	16.1 (15.1–17.4)	this study
		20	31.9	LT_{50}	–	this study
		20	31.8	CTM	–	Simons 1984
		20.5	33.4	CTM	–	Simons 1984

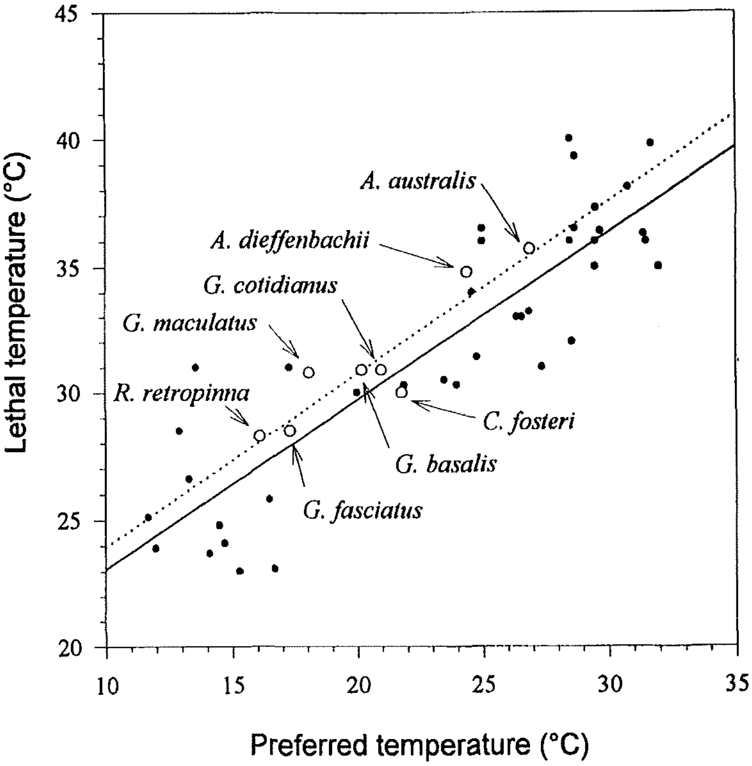

Fig. 3 Relationship between preferred and lethal temperatures for eight native fish species (open circles, dashed line), together with the regression line developed by Jobling (1981) for 38 other species (solid circles, solid line). See Table 2 for regression equations.

reported lethal temperatures. Boubée et al. (1991) showed that *G. maculatus* acclimated to 15°C began to avoid water of 23°C, and totally avoided temperatures higher than 29.5°C, which is only 1.3°C below the 10 minute LT_{50}.

For those genera that occur elsewhere, Sadler (1979) reported an ultimate upper lethal temperature of 38°C for the European eel (*A. anguilla*) which is comparable to the LT_{50} for *A. dieffenbachii* (37.3°C) and *A. australis* (39.7°C). The CTM of *R. retropinna* acclimated at 20°C (31.8°C) (Simons 1986), which was nearly identical to the LT_{50} (31.9°C), also coincided with that of a closely related Australian species, *R. semoni* (33.2°C) (Harasymiw 1983).

Many species exhibit an ontogenetic shift in thermal niche (Coutant 1987). For New Zealand's eels, lethal temperatures increased markedly with size, although the temperatures reported by Jellyman (1974) are anomalous because his trials were conducted over a 14 day period. This trend may be unusual among the genus *Anguilla*; Haro (1991) reported that glass eels of *A. rostrata* selected temperatures between 20 and 25°C, but that adult eels preferred lower temperatures between 17 and 20°C. Tesch (1977) also suggested that elvers are more sensitive than glass eels to changes in temperature. Although we did not determine the preferred temperature of large adult eels, Todd (1981) found shortest maturation times for male

Table 2 Linear regression analysis between preferred (X) and lethal (Y) temperatures of fish.

	n	Regression equation	*r*
Jobling (1981)	38	Y = 0.66X + 16.45	0.880
With New Zealand species	46	Y = 0.65X + 16.84	0.878
New Zealand species only	8	Y = 0.68X + 17.11	0.924

A. australis and *A. dieffenbachii* occurred at 26°C and 24°C respectively, which corresponds almost exactly to the preferred temperatures we determined for elvers.

Virtually no ontogenetic shift in temperature was observed for the other species where more than one life stage was tested (*G. maculatus* and *G. fasciatus*). Both lethal and preferred temperatures for different life stages tested were within a 1 to 2°C range (Table 1).

The fact that the addition of data for New Zealand species made little difference to Jobling's (1981) equation, suggests lethal temperatures could be used to predict preferred temperatures and vice versa. Jobling (1981) also described a linear relationship between the preferred or lethal temperature and the growth optima. Although we have no growth optima information for New Zealand species to compare with Jobling's (1981) equations, because New Zealand species demonstrate a similar relationship between lethal and preferred temperatures, growth optima could also be predicted with some confidence, and thus be used to recommend temperature regimes for native freshwater fish.

For example, using Armour's (1991) equations for calculating thermal regimes for protecting fish with data from this study, a maximum weekly average temperature for adult *R. retropinna* would be approximately 21.3°C. However, it must be remembered that life phases such as gonad development or spawning may have different thermal optima (Crawshaw 1977). Mora & Boubée (1993) showed that for *R. retropinna* egg development, highest survival and fewer larval deformities occurred at temperatures between 14 and 18°C, and that temperatures of 25°C and above caused a higher proportion of eggs to die. Again using Armour's (1991) equations, the maximum weekly average temperature for *R. retropinna* should decrease to around 19°C during the spawning season.

Several studies have demonstrated that temperature is important in determining the distribution and abundance of stream fish species (Matthews & Hill 1979; Moyle & Nichols 1982; Baltz et al. 1987; Brown 1989), supporting the assumption that fish seek out temperatures which match the most efficient operation of their metabolism (Coutant 1987). We used electric fishing field records from a database with national coverage (McDowall & Richardson 1983) to test this hypothesis for New Zealand species.

Each field record consisted of the latitude and altitude of the sampling site and the species that were present. A water temperature value was calculated for each site using latitude and altitude (Mosley 1982). The mean calculated temperature of all sites within the known latitude and altitude range for each species was compared to the mean calculated temperature of the sites where that species occurred to see if species were actually selecting sites where water temperatures matched that of their laboratory determined preferred temperatures (Table 3).

Three species did show a good relationship; *G. maculatus*, *G. fasciatus*, and *R. retropinna* inhabited sites with mean temperatures within about 1°C of their preferred temperatures. These species also had the lowest preferred temperatures of the eight species tested. On the other hand, eels, which had the highest preferred temperatures, were present at sites where mean temperatures were about 10°C below their preferred temperatures. The remaining species were found in localities where water temperature was about 4 to 6°C lower. These results are somewhat anomalous considering that other studies have shown laboratory determined temperature responses of fish correlate well with field results (Neill & Magnuson 1974; Matthews & Maness 1979; Matthews & Styron 1981; Brown 1989).

The relationship between fish density (no. m^{-2}), determined from the field records, and the mean calculated temperature was also examined. Only two species, *A. dieffenbachii* and *C. fosteri*, exhibited a significant relationship ($P < 0.05$) between density and temperature. Both results confirm Reynolds' (1977) statement that while species do have measurable optimal temperatures, many other factors, such as food, social and biotic interactions, stresses, disease etc., affect their ability to survive and grow, and may override or influence thermal response.

Our experiments determined the lethal and preferred temperatures for some common freshwater fish species, and showed that Jobling's (1981) equations, or the equation for just the New Zealand species, could be used to accurately predict lethal and preferred temperature, and by inference growth optima, from either one of these measures. While we would encourage New Zealand water managers to use this information, together with methods recommended by Armour (1991), to set temperature regimes for freshwater fish, it must be recognized that many factors affect fish distribution and

Table 3 Results from two-tailed Mann-Whitney comparison between the temperatures of all sites within the known altitude and latitude range for each species, and actual sites where each species was recorded. (* $P < 0.001$).

Species	*n*	Mean temperature ±SD (°C)	Preferred temperature (°C)
Anguilla australis	346	16.3 ± 1.8*	26.9
All sites	1049	15.1 ± 2.2	
A. dieffenbachii	851	14.9 ± 2.3*	24.4
All sites	1324	14.2 ± 2.8	
Gobiomorphus cotidianus	273	16.2 ± 2.1*	20.2
All sites	1324	14.2 ± 2.8	
G. basalis	87	15.7 ± 1.6	21.0
All sites	526	16.0 ± 1.8	
Galaxias maculatus	178	17.1 ± 1.8*	18.1
All sites	819	15.7 ± 2.0	
Cheimarrichthys fosteri	182	15.6 ± 2.1*	21.8
All sites	1116	14.9 ± 2.3	
G. fasciatus	153	16.1 ± 1.9*	17.3
All sites	1104	15.0 ± 2.3	
Retropinna retropinna	60	17.0 ± 1.7*	16.1
All sites	870	15.6 ± 2.1	

density. More research is needed to determine thermal requirements, especially avoidance temperatures, of critical life stages of most New Zealand species. In addition, water managers must balance the needs of individual species with those of the whole fish community; an ideal regime for *R. retropinna*, for example, may not satisfy that of an eel. However, New Zealand's freshwater fish appear to thrive in a wide range of temperatures, and water managers could go some way toward protecting the freshwater fish resource by ensuring lethal temperatures are avoided.

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